



**MBARARA UNIVERSITY OF SCIENCE AND
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COMPUTER SCIENCE**

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Course: Database Systems

Course code: CSC1204

Lecturer: Mr. Baguma Kenneth

Assignment: Database Project 1

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Discussion Group: 12

PROJECT REPORT

GROUP CHOSEN AREA OF INTEREST: AGRICULTURE

PROJECT TITLE: Farm Machinery and Equipment Management
Database System

DATABASE PROJECT

Agriculture – Farm Machinery and Equipment Management

This area focuses on managing;

- Tractors, ploughs, harvesters, basic farm tools among others.
- Equipment usage schedules
- Maintenance records
- Fuel consumption
- Operator assignments

Chosen System: Farm Machinery and Equipment Management Database System in the agriculture field

Mission of the project

To design and implement a reliable, efficient, and scalable database system that facilitates effective management, monitoring, and analysis of agricultural machinery and farm equipment to improve productivity and decision-making.

General Objective

To develop a centralized database system for managing agricultural machinery and farm equipment.

Specific Objectives

- To store and organize detailed information about farm machinery and equipment
- To track equipment availability, allocation, and operational status
- To manage maintenance schedules and repair records
- To control user access and system security
- To improve efficiency in data retrieval and management
- To generate reports for informed decision-making
- To monitor costs associated with equipment usage and maintenance

- To ensure data integrity and consistency

Reasons:

Machinery is central to modern agriculture productivity therefore highly relevant in the agricultural field

The broad scope of machinery combines inventory, maintenance, and usage tracking of farm tools hence highly used.

The database system addresses problems like;

- ✓ Equipment misuse
- ✓ Poor maintenance tracking thus leads to the loss of machinery
- ✓ Farms can later integrate with advanced tools and records

Functions of the database system

- ◇ Registering all farm equipment
- ◇ Tracking equipment usage (who used it, when and for what task)
- ◇ Scheduling maintenance and repairs
- ◇ Monitoring availability and allocation
- ◇ Recording fuel consumption
- ◇ Generating reports for management

User Groups

- Farm managers
- Equipment operators
- Maintenance technicians
- Agricultural officers / supervisors

System Benefits after implementation

- ❖ Systematic reduced maintenance costs
- ❖ Increased equipment lifespan
- ❖ Improved utilization of machinery

- ❖ Reduced fuel wastage
- ❖ Better planning of farm operations
- ❖ Improved decision-making through data insights
- ❖ Increased accountability of workers
- ❖ Better record keeping and transparency
- ❖ Enhanced productivity and efficiency
- ❖ Scalability is maximised thus easy retrieval of records

Analysis of Costs

- ◆ Tangible Costs:
 - ◆ Hardware (computers, servers)
 - ◆ Software development
 - ◆ Training costs
 - ◆ Maintenance and updates

Expected hardships

- Resistance to change from staff
- Time spent learning the system
- Temporary productivity drop during transition

Benefits outweigh costs, especially long-term (cost savings + efficiency gains)

User Issues

- ❖ Skill Levels:
 - ❖ Some users may have limited ICT skills
- ❖ Training Needs:
 - ❖ Basic computer and system usage training required
- ❖ Adoption Challenges:
 - ❖ Resistance from workers used to manual systems

Mitigation to the user issues

- ❖ Simple User Interface thus easy to adapt
- ❖ Conduct training sessions
- ❖ Offer user manuals and support

Risk Assessment

Technical Risks	Mitigation
➤ System failure or bugs ➤ Data loss or corruption ➤ Poor system design	➤ Regular backups ➤ Testing before deployment
Organizational Risks	Mitigation
➤ Resistance to adoption ➤ Lack of management support	➤ Stakeholder involvement ➤ Awareness and training
Operational Risks	Mitigation
➤ Incorrect data entry ➤ Misuse of system	➤ Validation rules ➤ User access control

Limitations of the System

May not support real-time tracking without IoT integration

Depends on user input accuracy

Limited functionality in areas with poor electricity or internet

Initial setup cost may be high for small-scale farmers

Does not fully automate farm operations (only supports management)

PROJECT ENTITIES AND THEIR ATTRIBUTES

1. Equipment

equipment_id (PK)
equipment_name
category_id (FK)
category_name
status

equipment_id (FK)
user_id (FK)
start_date
end_date
task_name

2. Branch

branch_id(PK)
manager
branch_name

6. Maintenance

maintenance_id (PK)
equipment_id (FK)
user_id (FK)
date
description
amount

3. User

user_id (PK)
name
email
phone
qualification
location
role_id
role_name

7. FuelLog

fuel_id (PK)
equipment_id (FK)
date
quantity
cost
userName(FK)

4. Task

task_id (PK)
task_name

8. Expense

expense_id (PK)
equipment_id (FK)
expense_category
amount
date

5. Deployment

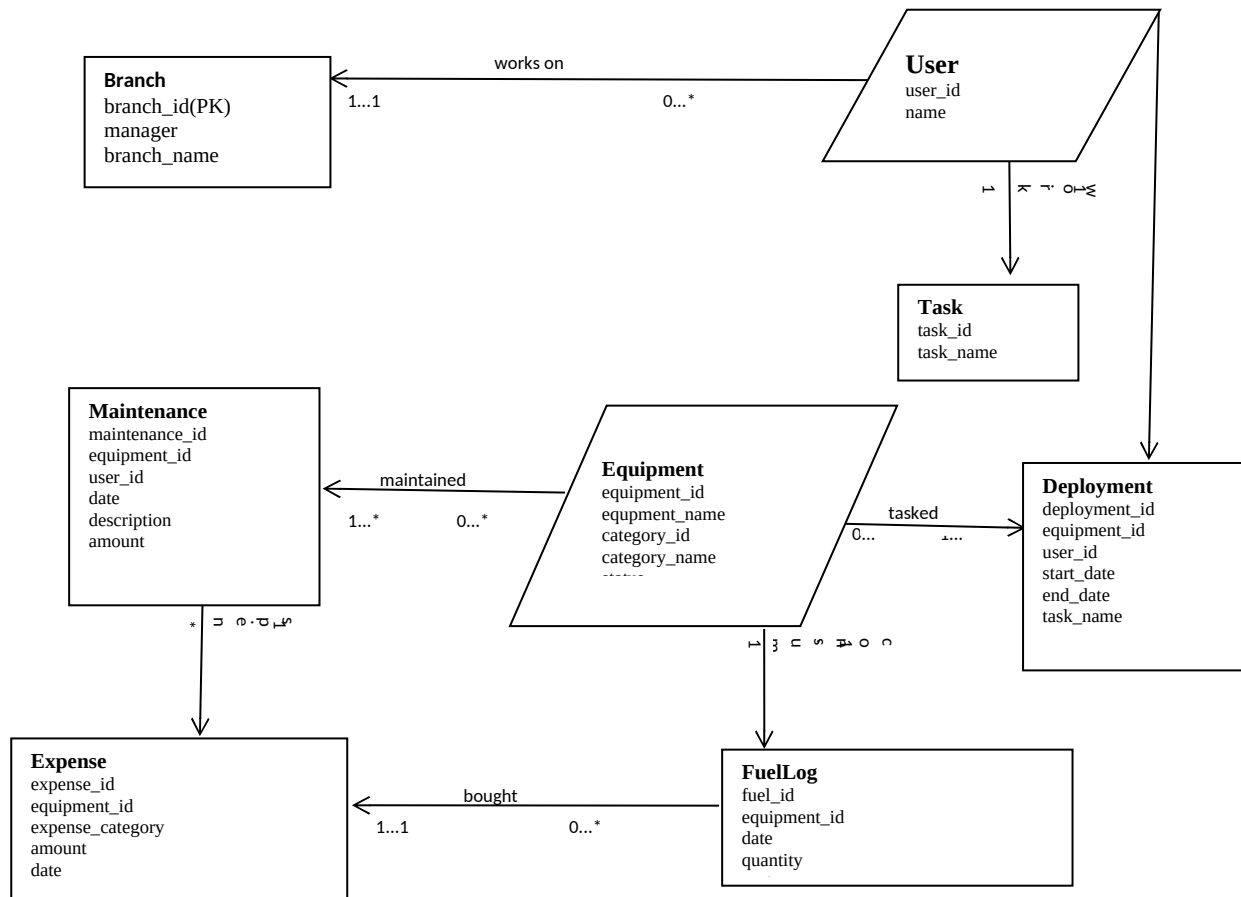
deployment_id (PK)

Binary Relationships between entities

Equipment → Branch (Many-to-One)
Equipment → User (One-to-Many)
Equipment → Maintenance (zero-to-many)
Equipment → Deployment (zero-to-many)

Equipment → FuelLog (One-to-Many)
User → equipment (zero-to-many)
User → Task (Many-to-One)
User → Deployment (zero-to-many)
User → Maintenance (zero-to-many)
FuelLog → Expense (one-to-one)
User → Branch (one-to-one)
Maintainence → Expense (one-to-many)

ENTITY RELATIONAL DIAGRAM



THE PHYSICAL SCHEMA

I created the database and named it FarmMachineryAndEquipmentManagement

```
mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| BCS2025  |
| FarmMachineryAndEquipmentManagement |
| bcs2025  |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
7 rows in set (0.00 sec)
```

I inserted the tables into the database and their Attributes.

Using the command,

INSERT TABLE tableName(Attribute1,attribute2,...);

I inserted the records into the attributes using commands like;

```
mysql> INSERT INTO user VALUES("U0001", "Kwesiga Devine", "kwesigadevine001@gmail.com", 0702378436, "Diploma mechanics", "Masaka", "R001", "Manager");
Query OK, 1 row affected (0.18 sec)

mysql> INSERT INTO user VALUES("U0002", "Tusiime Maria Desire", "tusiimemariad001@gmail.com", 0776332126, "BSc Agriculture", "Mbarara", "R003", "Supervisor");
Query OK, 1 row affected (0.43 sec)

mysql> INSERT INTO user VALUES("U0002", "Tusiime Maria Desire", "tusiimemariad001@gmail.com", 0776332126, "BSc Agriculture", "Mbarara", "R003", "Supervisor");
Query OK, 1 row affected (0.14 sec)

mysql> INSERT INTO user VALUES("U0005", "Kabunga Charles", "kabungacharles001@gmail.com", 0727642126, "MSc Engineering", "Masaka", "R005", "Operator");
Query OK, 1 row affected (0.48 sec)

mysql> INSERT INTO user VALUES("U0004", "Mubbiru Clever", "mubbiruclever@gmail.com", 0754674352, "Diploma Agronomy", "Entebbe", "R004", "Technician");
Query OK, 1 row affected (0.44 sec)
```

After successful INSERTION of the tables in my database,

```
mysql> SHOW TABLES;
+-----+
| Tables_in_FarmMachineryAndEquipmentManagement |
+-----+
| branch |
| deployment |
| equipment |
| expense |
| fuelLog |
| maintenance |
| task |
| user |
+-----+
8 rows in set (0.01 sec)
```

After inserting values in the attributes of the database,

I queried to see the table content using the **SELECT** command.

Task entity

```
mysql> SELECT * FROM task;
```

taskId	taskName
1	Cultivation
2	Maintenance
3	Weeding
4	Spraying
5	Destructuring

```
5 rows in set (0.00 sec)
```

User entity

```
mysql> SELECT * FROM user;
```

userId	name	email	phone	qualification	location	roleId	roleName
U0001	Kwesiga Devine	kwesigadevine001@gmail.com	702378436	Diploma mechanics	Masaka	R001	Manager
U0005	Kabunga Charles	kabungacharles001@gmail.com	727642126	MSc Engineering	Masaka	R005	Operator
U0004	Mubbiru Clever	mubbiruclever@gmail.com	754674352	Diploma Agronomy	Entebbe	R004	Technician
U0003	Tumwine David	tumwiinedavid111@gmail.com	743563487	BSc Farming	Kamwokya	R002	Admin
U0002	Tusiime Maria Desire	tusiimemariad001@gmail.com	776332126	BSc Agriculture	Mbarara	R003	Supervisor

```
5 rows in set (0.00 sec)
```

Maintenance entity

```
mysql> SELECT * FROM maintenance;
```

maintenanceId	equipmentId	userId	date	description	amount	endDate	taskName
M0001	E0002	U0001	20-03-2026	Welding rod replace	150000	21-03-2026	Replacement
M0002	E0004	U0002	24-03-2026	Blade shapping	150000	25-03-2026	Sharpening
M0003	E0005	E0004	22-03-2026	Drill bit change	110000	22-03-2026	Replacement
M0004	E0006	U0003	25-03-2026	Handle tightening	30000	25-03-2026	Repair
M0005	E0003	U0005	25-03-2026	chain Oiling	90000	25-03-2026	Lubrication
M0006	E0007	U0004	28-03-2026	Cultivation	1700000	02-03-2026	Cultivation

```
6 rows in set (0.00 sec)
```

FuelLog entity

```
mysql> SELECT * FROM fuelLog;
```

fuelId	equipmentId	date	quantity	cost	userName
FL001	E0001	22/04/2026	10	66500	Mubbiru Clever
FL002	E0003	20/04/2026	10	66500	Kwesiga Devine
FL003	E0005	25/04/2026	4	31500	Kabunga Charles
FL004	E0004	29/04/2026	20	140000	Tusiime Maria Desire

```
4 rows in set (0.00 sec)
```

Expense entity

```
mysql> select * from expense;
```

expenseId	equipmentId	expenseCategory	amount	date	userId
EX001	E0001	Fuel	70000	20-03-2026	U0001
EX002	E0002	Maintenance	150000	22-03-2026	U0002
EX003	E0003	Lubrication	80000	25-03-2026	U0003
EX004	E0004	Maintenance	150000	24-03-2026	U0004
EX004	E0005	Maintenance	90000	28-03-2026	U0004
EX006	E0006	Repair	140000	25-03-2026	U0006
EX007	E0007	Fuel	30000	29-03-2026	U0007
EX008	E0008	Spare Parts	75000	30-03-2026	U0008

```
8 rows in set (0.00 sec)
```

Equipment entity

```
mysql> SELECT * FROM equipment;;
```

equipmentId	equipmentName	categoryId	categoryName	status
EQ001	Lawn mower	CAT01	Garden Equipments	Needs a bigger grass collector
EQ002	Welding Machine	CAT02	Maintenance Tools	Works very well
EQ003	Chain Saw	CAT03	Garden Equipment	Needs a sharper blade
EQ004	Ox Plow	CAT04	Garden Equipments	Requires a stronger carrier
EQ005	Electric Drill	CAT05	Power Tools	Very efficient
EQ006	Hammer	CAT06	Hand Tools	Works well though not firm
EQ007	Tractor	CAT07	Garden Equipment	Works well but needs gears

```
7 rows in set (0.00 sec)
```

Deployment entity

```
mysql> SELECT * FROM deployment;
```

deploymentId	equipmentId	userId	startDate	endDate	taskName
DP004	EQ001	U0004	20/04/2026	25/04/2026	Weeding
DP002	EQ005	U0001	24/04/2026	28/04/2026	Maintenance
DP003	EQ005	U0002	23/04/2026	27/04/2026	Cultivation
DP001	EQ003	U0003	22/04/2026	23/04/2026	Spraying
DP005	EQ002	U0005	28/04/2026	4/05/2026	Repairing Tractor

```
5 rows in set (0.00 sec)
```

Branch entity

```
mysql> SELECT * FROM branch;
```

branchId	manager	branchName
1	Kabunga Charles	Mbarara
2	Tusiime Maria Desire	Kampala
3	Kwesiga Devine	Lukaya
4	Nakalema Betty	Masaka
5	Mubiru Clever	Kirumba

```
5 rows in set (0.01 sec)
```

QUERY COMMANDING USING DML ON THE LOGICAL SCHEMA

I realised i had forgotten to add a column in expense table and i used the **ALTER TABLE 'ADD COLUMN'** command to add **userId** as both tables have a relationship and **userId** will referebnce user table in the expense table helping in easy report making

```
mysql> ALTER TABLE expense ADD COLUMN userId VARCHAR(5);  
Query OK, 0 rows affected (0.37 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

I used the **UPDATE** command to update record values misentered. Thus easy management due to data independency

```
mysql> UPDATE expense SET userId="U0002" WHERE expenseId="EX002";
Query OK, 1 row affected (0.13 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE expense SET userId="U0003" WHERE expenseId="EX003";
Query OK, 1 row affected (0.20 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE expense SET userId="U0004" WHERE expenseId="EX004";
Query OK, 1 row affected (0.12 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE expense SET userId="U0004" WHERE expenseId="EX005";
Query OK, 1 row affected (0.25 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

In some tables, i had not set the primary key and set them using **ALTER TABLE ... 'ADD PRIMARY KEY' ..** command

```
mysql> ALTER TABLE fuelLog ADD PRIMARY KEY (fuelId);
Query OK, 0 rows affected (1.99 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE deployment ADD PRIMARY KEY (deploymentId);
Query OK, 0 rows affected (1.49 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> ALTER TABLE branch ADD PRIMARY KEY (branchId);
Query OK, 0 rows affected (1.97 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Selecting only users located in Mbarara easing task allocation

```
mysql> SELECT * FROM user WHERE location = "Mbarara";
+-----+-----+-----+-----+-----+-----+-----+-----+
| userId | name           | email                      | phone | qualification | location | roleId | roleName |
+-----+-----+-----+-----+-----+-----+-----+-----+
| U0002  | Tuslime Maria Desire | tuslimemariad001@gmail.com | 776332126 | BSc Agriculture | Mbarara | R003   | Supervisor |
+-----+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Selecting records in expense table whose amount is greater than 100000

This eases easy filtering of fields with a higher expense for better planning

```
mysql> SELECT * FROM expense WHERE amount > 100000;
+-----+-----+-----+-----+-----+-----+
| expenseId | equipmentId | expenseCategory | amount | date       | userId |
+-----+-----+-----+-----+-----+-----+
| EX002     | EQ002       | Maintenance     | 150000 | 22-03-2026 | U0002  |
| EX004     | EQ004       | Maintenance     | 150000 | 24-03-2026 | U0004  |
| EX006     | EQ006       | Repair          | 140000 | 25-03-2026 | U0006  |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Using relational calculus **JOIN** command, i retrieved records in some attributes from **maintenance** and **user** tables with the same userId.

This enables quick report making by knowing the operators and the respective tasks handled.

```
mysql> SELECT m.maintenanceId, u.name, m.description, m.amount, m.date FROM maintenance m JOIN user u ON m.userId = u.userId;
```

maintenanceId	name	description	amount	date
M0001	Kwesiga Devine	Welding rod replace	150000	20-03-2026
M0002	Tusiime Maria Desire	Blade shapping	150000	24-03-2026
M0004	Tumwiine David	Handle tightening	30000	25-03-2026
M0005	Kabunga Charles	chain Oiling	90000	25-03-2026
M0006	Mubbiru Clever	Cultivation	1700000	28-03-2026

5 rows in set (0.00 sec)

I ordered the records in maintenance table by date in ascending order, this helps a supervisor to quickly know the sheduled tasks in their order and accomplished tasks for easy maintainability

```
mysql> SELECT * FROM maintenance ORDER BY date ASC;
```

maintenanceId	equipmentId	userId	date	description	amount	endDate	taskName
M0001	EQ002	U0001	20-03-2026	Welding rod replace	150000	21-03-2026	Replacement
M0003	EQ005	EQ004	22-03-2026	Drill bit change	110000	22-03-2026	Replacement
M0002	EQ004	U0002	24-03-2026	Blade shapping	150000	25-03-2026	Sharpening
M0004	EQ006	U0003	25-03-2026	Handle tightening	30000	25-03-2026	Repair
M0005	EQ003	U0005	25-03-2026	chain Oiling	90000	25-03-2026	Lubrication
M0006	EQ007	U0004	28-03-2026	Cultivation	1700000	02-03-2026	Cultivation

6 rows in set (0.00 sec)

Conclusion

Farm Machinery and Equipment Management Database System provides a structured and the best efficient approach to managing agricultural machinery. By improving equipment tracking, maintenance scheduling, and operational planning, the system enhances productivity and reduces operational costs. Despite challenges such as user adoption and maintenance costs, the long-term benefits significantly justify its implementation, making it a viable and impactful solution in the agricultural sector.

Database Administrator and Designer: Kabunga Charles